Housing Eviction Research Proposal [Team One]

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Problem And Goal

Question: Do eviction filing rates differ between Cook County estimates with many housing-voucher households and those with few, after accounting for income and rent burden, from 2015 to the latest year available?

- This question matters because eviction can lead to job loss, homelessness, and worse health and school outcomes. Housing staff and legal aid groups need to know where to focus limited resources.
- We would be using the Rashida048 guideline (Learn to Formulate Good Research Question For Efficient Statistical Analysis).





Our Data Plan

Step 1: Look through the README

The archive has four groups of files:

Eviction data: 4 CSVs, one row per year (2010 to 2019) for each area.

Census data: 4 CSVs, American Community Survey 5-year estimates for 2006-2010 and 2014-2018.

Crosswalks: 3 CSVs that map census tracts to community areas, wards, and Chicago.

Documentation: the methodology and field-definition PDFs.

The eviction data covers residential cases filed in Chicago. Cook County cases are reviewed but not included here.

Step 2: Merge the relevant information

Each eviction file pairs with one census file and one crosswalk:

eviction_data_chicago.csv with census_data_chicago.csv

eviction_data_comm_area.csv with census_data_comm_area.csv and census_tract_crosswalk_comm_area.csv

eviction_data_ward.csv with census_data_ward.csv and census_tract_crosswalk_ward.csv

eviction_data_tract.csv with census_data_tract.csv

Step 3: Explore all of the data broken down into census data in Chicago, community data, wards and census tracts.

Geographies:

Community areas: 77
Wards: 50
Census tracts: 804

Community areas, 2019:

Most filings: South Shore -> 1,741
Highest rate: South Shore -> 8.72
Lowest rate: Lincoln Park -> 0.47

Census, citywide:

Median rent, 2006-2010 to 2014-2018: $885 to $1,077
Rental units:
592,752 to 626,929

Channels

Where the data is published and how it is cited.

Portal: eviction.lcbh.org, run by the Lawyers' Committee for Better Housing.

Files: the CSVs and PDFs in this archive.

Citation: Chicago Evictions, Lawyers' Committee for Better Housing, Release 2, 17 December 2020.

Post-launch plan

Inspect the data, learn from it, and decide what specific data mining statistical processes we plan on doing.

Formalization Into Data Mining

Step 1

Combine all datasets in into one csv

Step 2

Remove rows(information) not useful for our project

Step 3

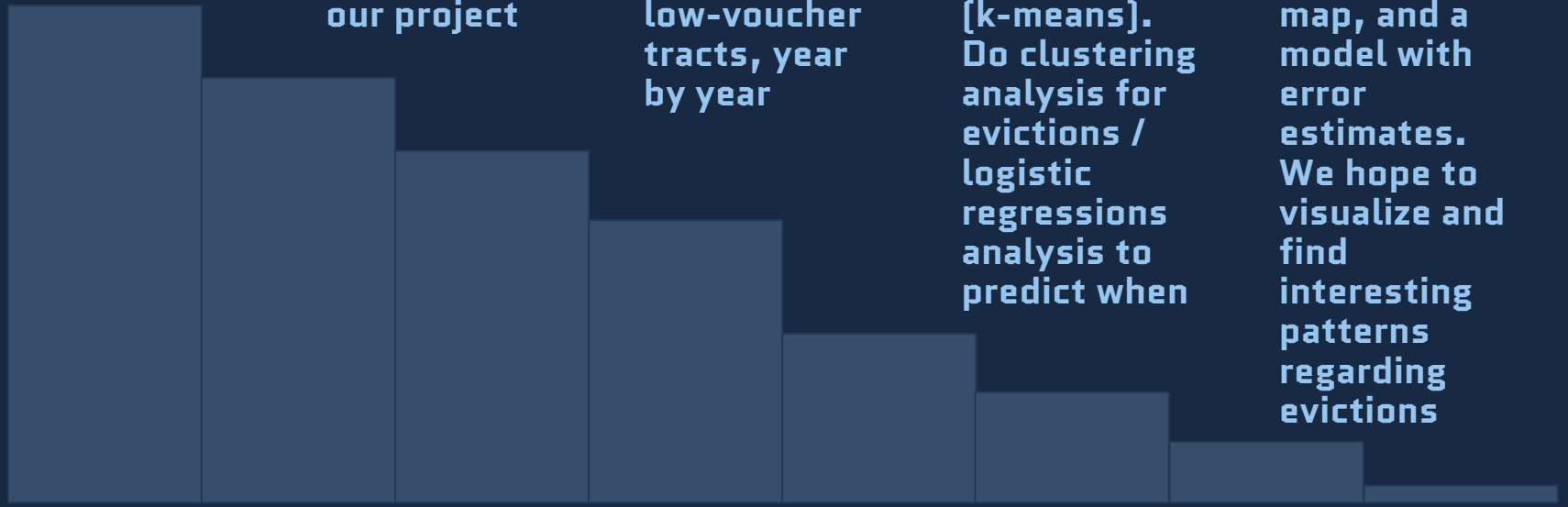
Compare filing rates between high- and low-voucher tracts, year by year

Step 4

Cluster tracts to find eviction hotspots (k-means). Do clustering analysis for evictions / logistic regressions analysis to predict when

Step 5

Make a table of rate differences, a hotspot map, and a model with error estimates. We hope to visualize and find interesting patterns regarding evictions



Teamwork Agreement

Team Contract

Our team contract follows the University of Maryland's teamwork agreement template.
Teamwork Agreement

Github Repository

Here is our GitHub Repo for all of our project code.
GithubRepoGroup1CS483
cs483spring2026project@gmail.com
Permission has already been granted to:
dmdh@uic.edu
hhao@uic.edu

Development Model

For this project, we will use the Agile software development model as described in clutch.co.

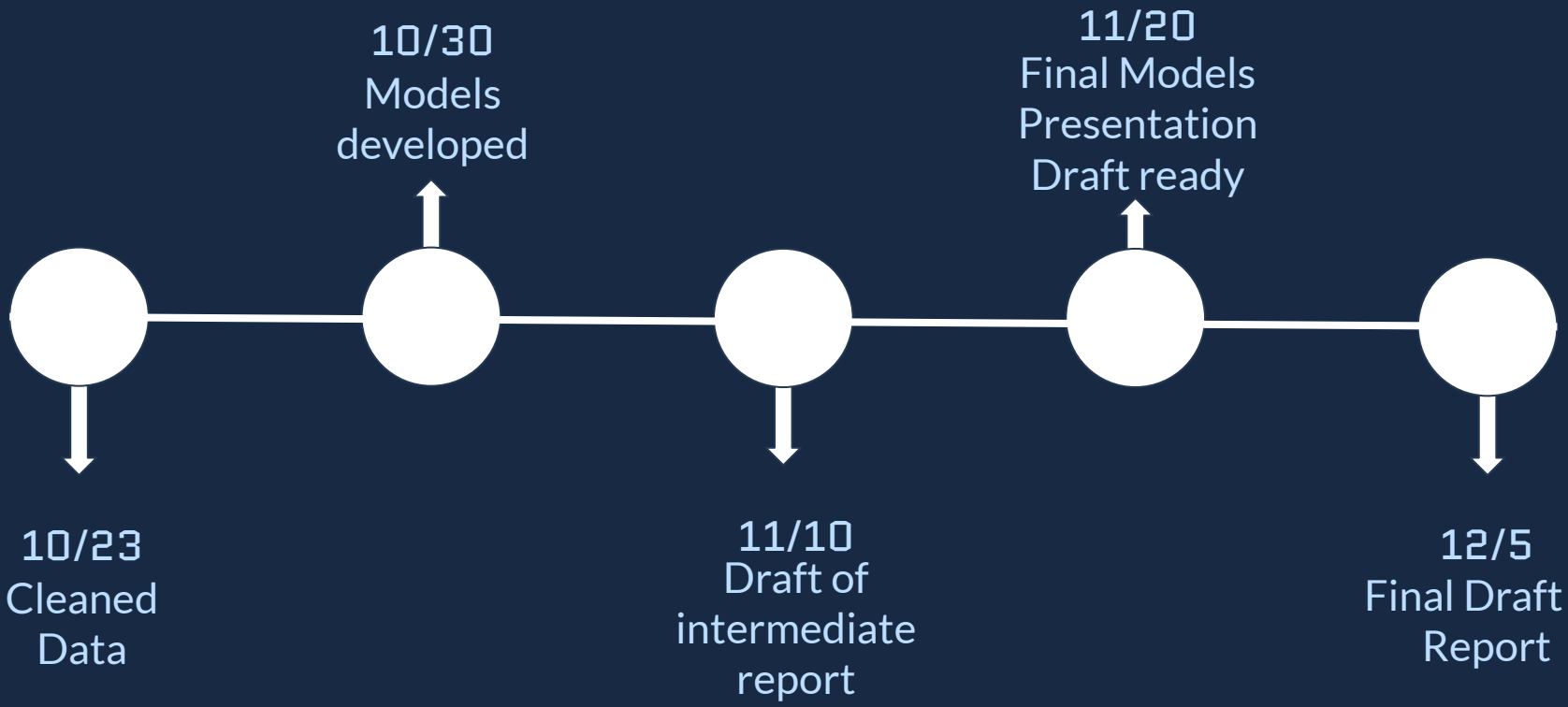
The Agile model has been selected because of its flexibility and will allow us to continue to practice and refine our SCRUM skills.

Project Management Tool: Jira

https://cs483spring2026housing.atlassian.net/jira/software/projects/SCRUM/boards

Note: the following two emails have been added to the Atlassian Jira Board:
dmdh@uic.edu
hhao@uic.edu

Our Schedule



Dataset Links Cited:

http://eviction.lcbh.org/
https://www.nytimes.com/interactive/2014/06/27/us/detroit-foreclosure-photo-mosaic.html
https://chicagohealthatlas.org/indicators/EVC?tab=chart
https://evictionresearch.net/illinois/

